

Proanthocyanins

Closely related to tannins and flavonoids, these polyphenolic compounds are pigments that give flowers and fruits a blue, purple, or red hue. They are powerfully antioxidant and free-radical scavengers. They protect the circulation from damage, especially the circulation in the heart, hands, feet, and eyes. Blackberry (*Rubus fruticosus*, p. 264), red grapes (*Vitis vinifera*, p. 283), and hawthorn (*Crataegus oxycantha*, p. 87) all contain appreciable quantities of these proanthocyanins.



Blackberry
(*Rubus fruticosus*)

Coumarins

Coumarins of different kinds are found in many plant species and have widely divergent actions. The coumarins in melilot (*Melilotus officinalis*, p. 234) and horse chestnut (*Aesculus hippocastanum*, p. 58) help to keep the blood thin, while furanocoumarins such as bergapten, found in celery (*Apium graveolens*, p. 64), stimulate skin tanning, and khellin, found in visnaga (*Ammi visnaga*, p. 62), is a powerful smooth-muscle relaxant.



Celery
(*Apium graveolens*)

Saponins

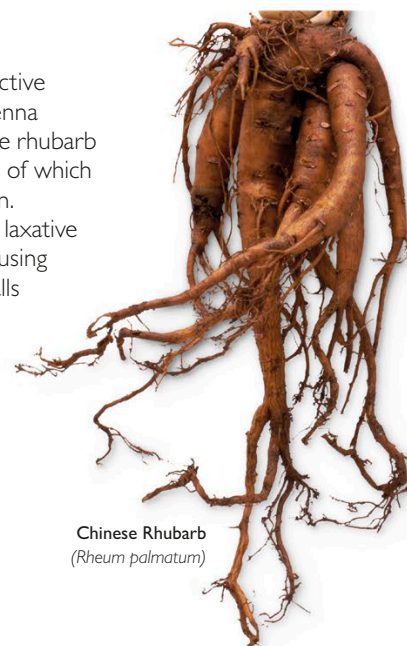
The main active constituents in many key medicinal plants, saponins gained their name because, like soap, they make a lather when placed in water. Saponins occur in two different forms—steroidal and triterpenoid. The chemical structure of steroidal saponins is similar to that of many of the body's hormones, for example estrogen and cortisol, and many plants containing them have a marked hormonal activity. Wild yam (*Dioscorea villosa*, p. 91), from which the contraceptive pill was first developed, contains steroidal saponins. Triterpenoid saponins occur more commonly—for example in licorice (*Glycyrrhiza glabra*, p. 101) and cowslip root (*Primula veris*, p. 256)—but have less hormonal activity. They are often expectorant and aid absorption of nutrients.



Licorice
(*Glycyrrhiza glabra*)

Anthraquinones

Anthraquinones are the main active constituents in herbs such as senna (*Cassia senna*, p. 75) and Chinese rhubarb (*Rheum palmatum*, p. 126), both of which are taken to relieve constipation. Anthraquinones have an irritant laxative effect on the large intestine, causing contractions of the intestinal walls and stimulating a bowel movement approximately 10 hours after being taken. They also make the stool more liquid, easing bowel movements.



Chinese Rhubarb
(*Rheum palmatum*)

Cardiac Glycosides

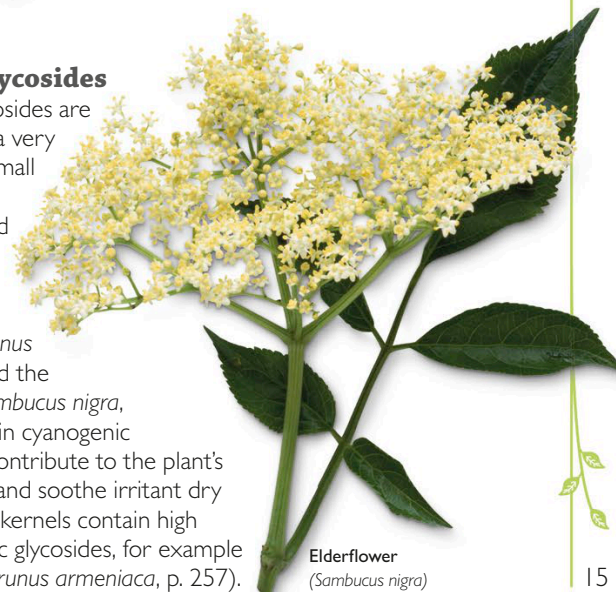
Found in various medicinal plants, notably in foxgloves (see common foxglove, *Digitalis purpurea*, p. 202) and in lily of the valley (*Convallaria majalis*, p. 194), cardiac glycosides such as digitoxin, digoxin, and convallotoxin have a strong, direct action on the heart, supporting its strength and rate of contraction when it is failing. Cardiac glycosides are also significantly diuretic. They help to stimulate urine production, thus increasing the removal of fluid from the tissues and circulatory system.



Common Foxglove
(*Digitalis purpurea*)

Cyanogenic glycosides

Though these glycosides are based on cyanide, a very potent poison, in small doses they have a helpful sedative and relaxant effect on the heart and muscles. The bark of wild cherry (*Prunus serotina*, p. 257) and the leaves of elder (*Sambucus nigra*, p. 132) both contain cyanogenic glycosides, which contribute to the plant's ability to suppress and soothe irritant dry coughs. Many fruit kernels contain high levels of cyanogenic glycosides, for example those of apricot (*Prunus armeniaca*, p. 257).



Elderflower
(*Sambucus nigra*)